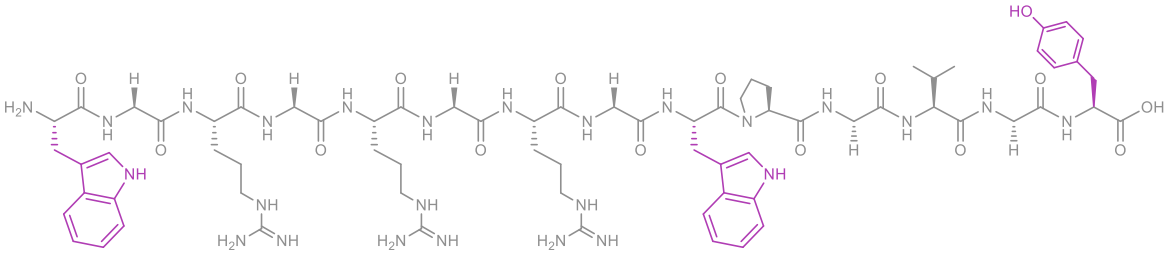
Supporting Information

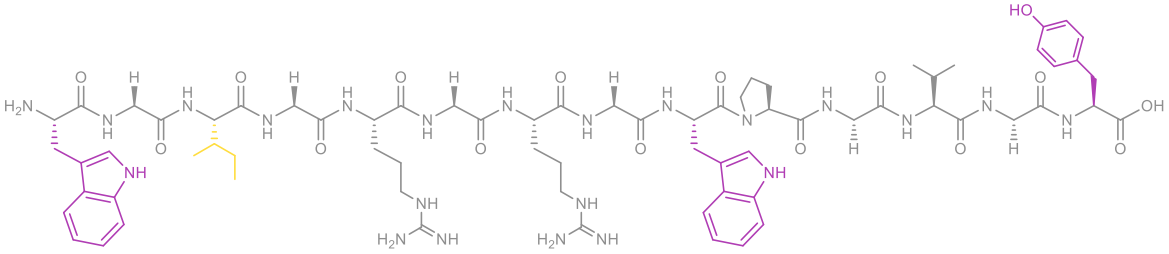
Regulation of Chemical Transformation in Designer Peptide Biomolecular Condensates

Shirel Veretnik[1], Avigail Baruch Leshem[1] and Ayala Lampel*[,1-4]

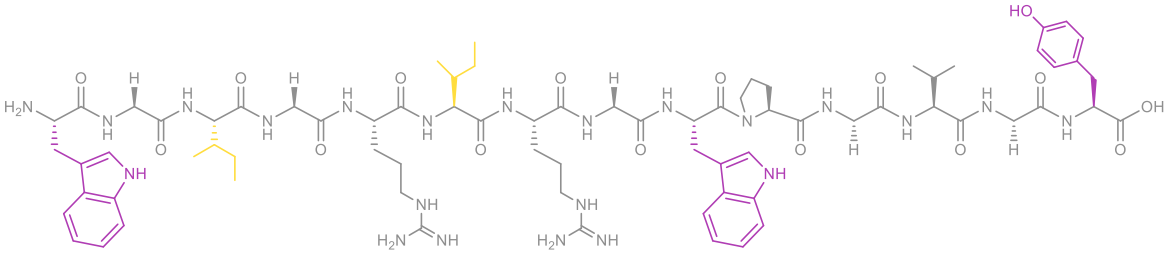
a.



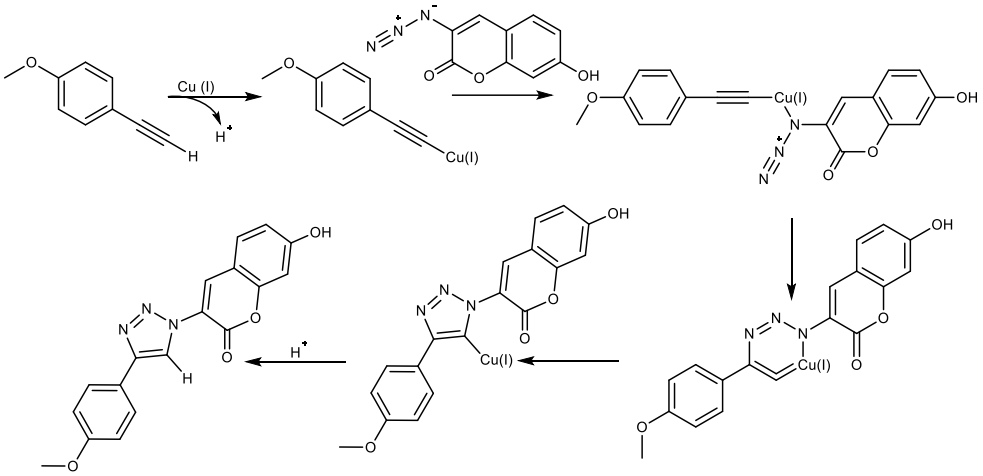
b.



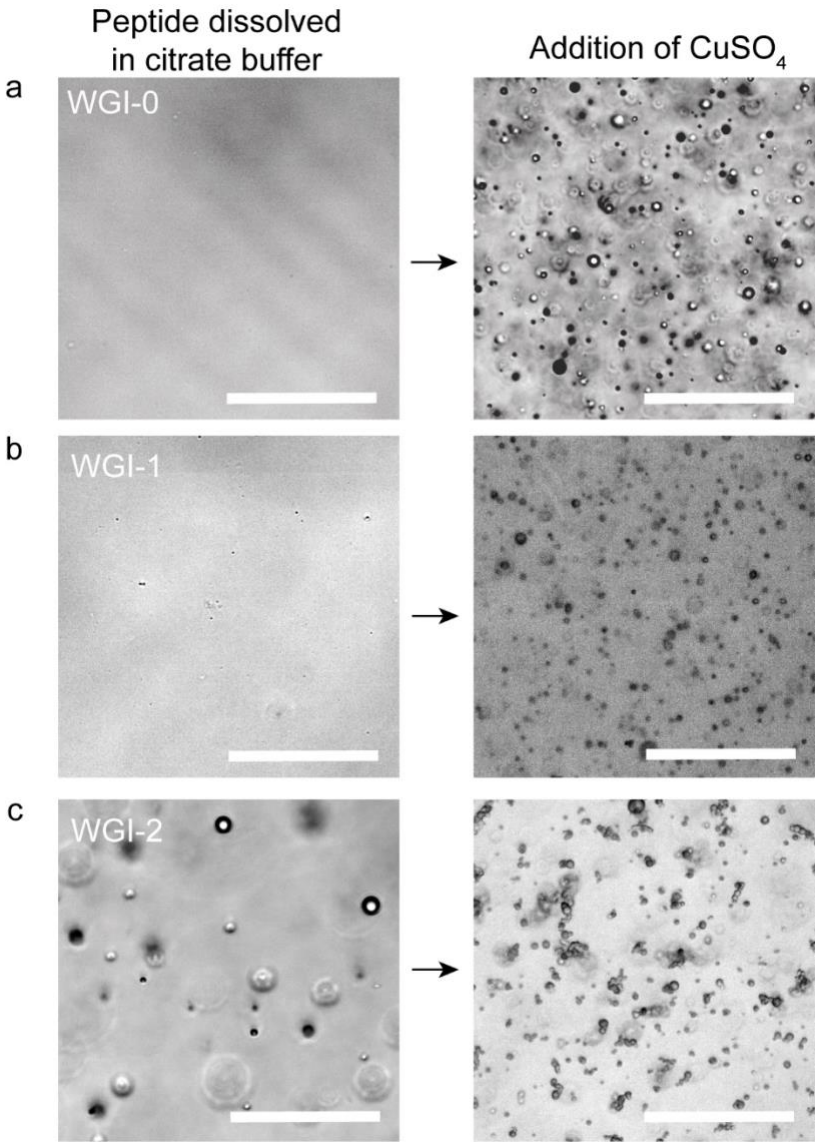
c.



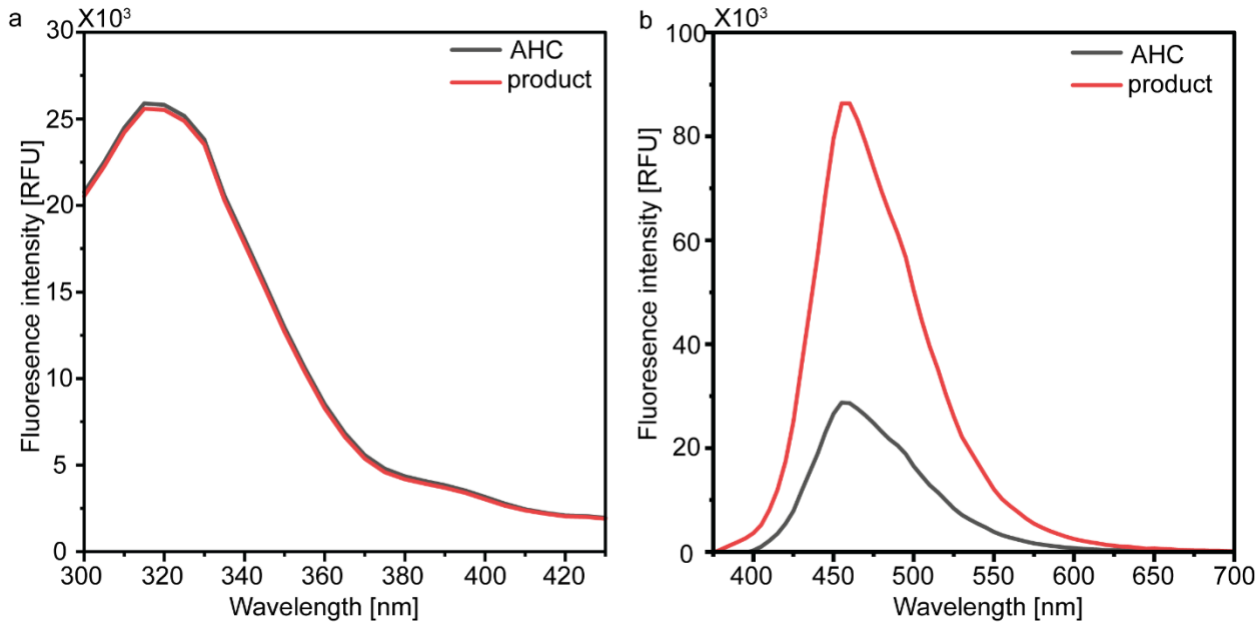
Supplementary Figure 1. Chemical structures of **a.** WGI-0, **b**. WGI-1, and **c**. WGI-2 peptides.



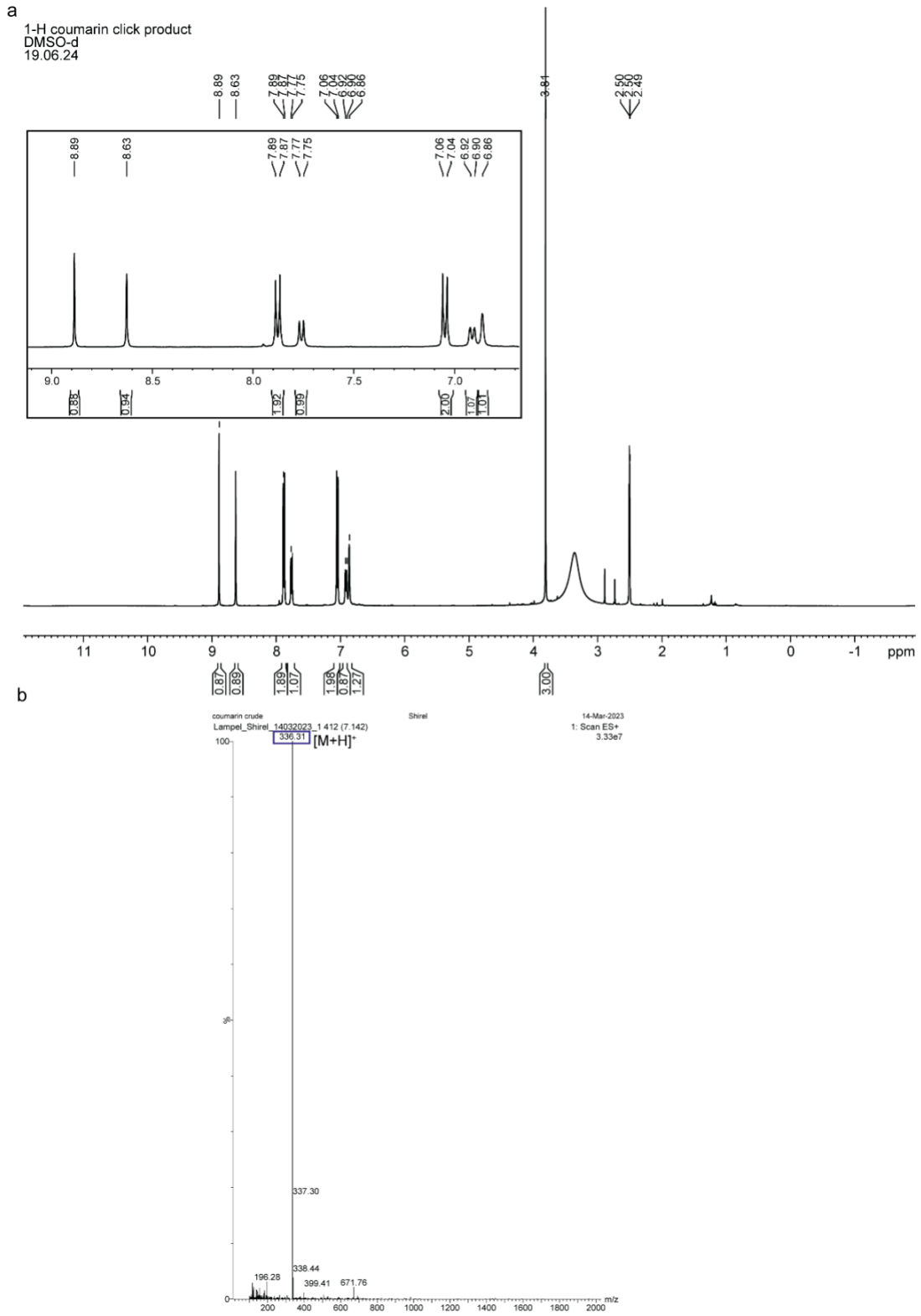
Supplementary Figure 2. Click reaction mechanism between 3-Azido-7-hydroxycoumarin (AHC) and 4-ethynylanisole to form 1,4-bis(4-methoxyphenyl)-1H-1,2,3-triazole.



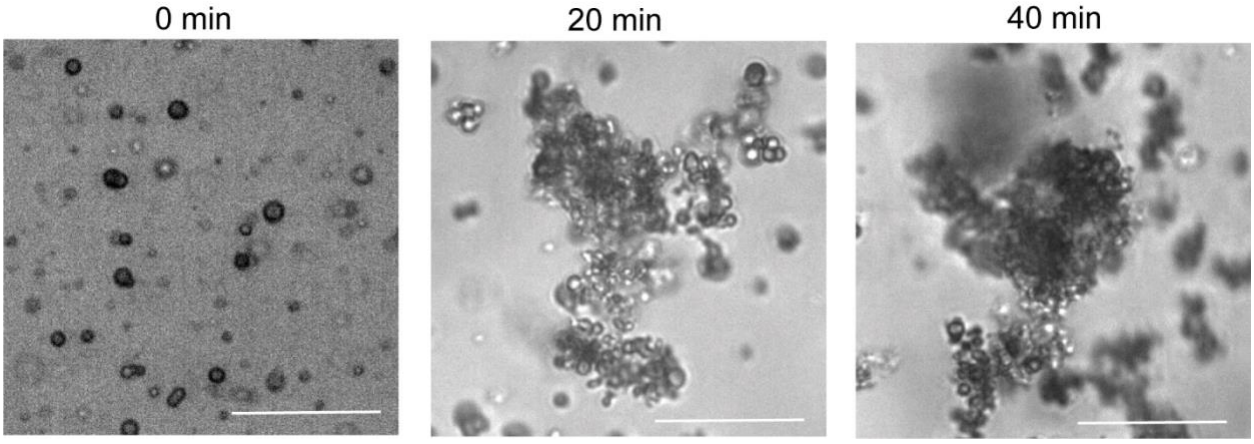
Supplementary Figure 3. Droplets formation by the addition of $CuSO_4$ to a 5 mM of **a.** WGI-0, **b.** WGI-1, and **c.** WGI-2. Peptides were dissolved in citrate buffer at pH 5.5, then $CuSO_4$ was added to the peptide solutions, resulting in condensate formation. Scale bar=50 µm.



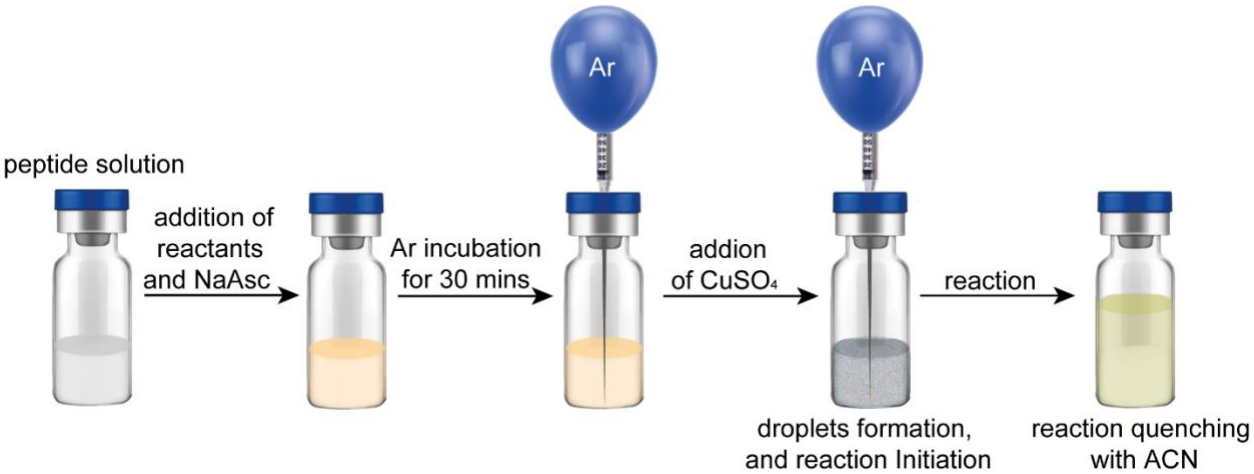
Supplementary Figure 4. a. Excitation spectra of product and AHC λ_{em}=330 nm. **b**. Emission spectra of product and AHC λ_{ex}=460 nm.



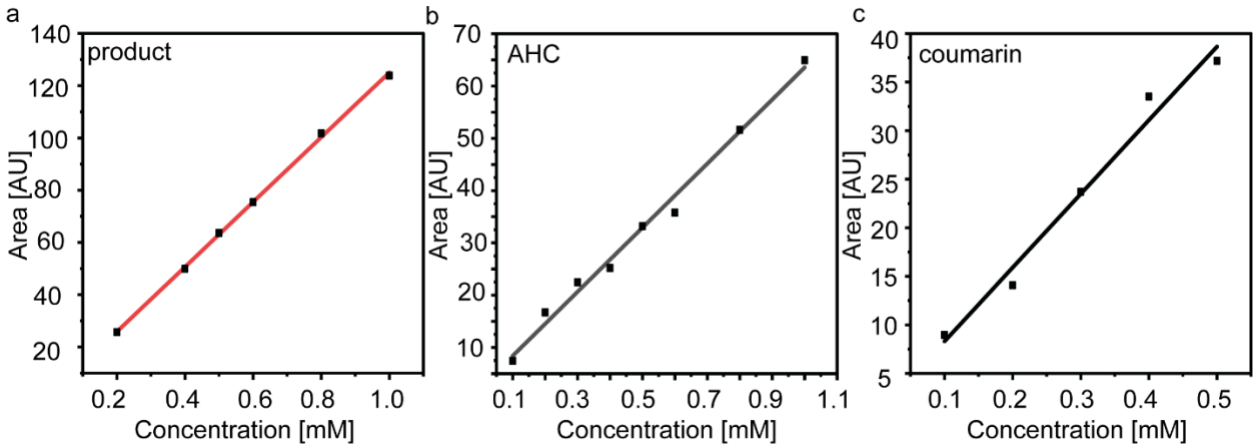
Supplementary Figure 5. a. H^1 NMR spectrum and **b.** MS spectrum of 7-hydroxy-3-(4-(4-methoxyphenyl)-1H-1,2,3-triazol-1-yl)-2H-chromen-2-one.



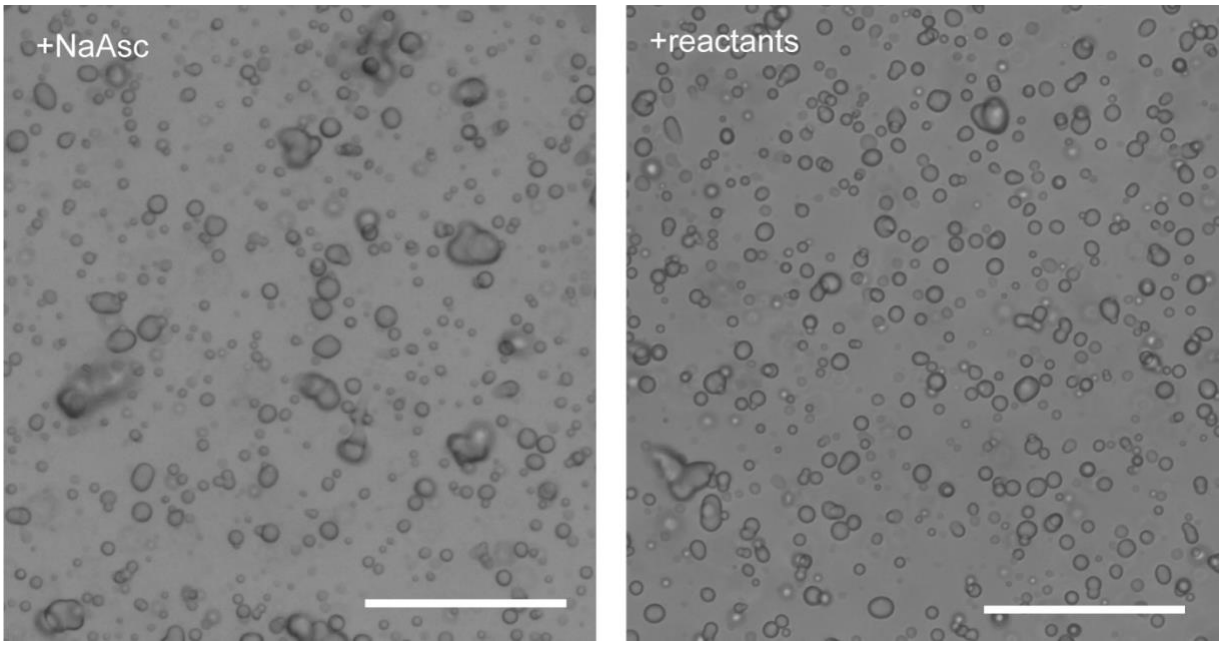
Supplementary Figure 6. 5 mM WGI-1 in citrate buffer, pH 5.5 with 1 mM reactants, 20 mM NaASc, and 10 mM CuSO4, at t=0, 20 and 40 min. Scale bar=50 µm.



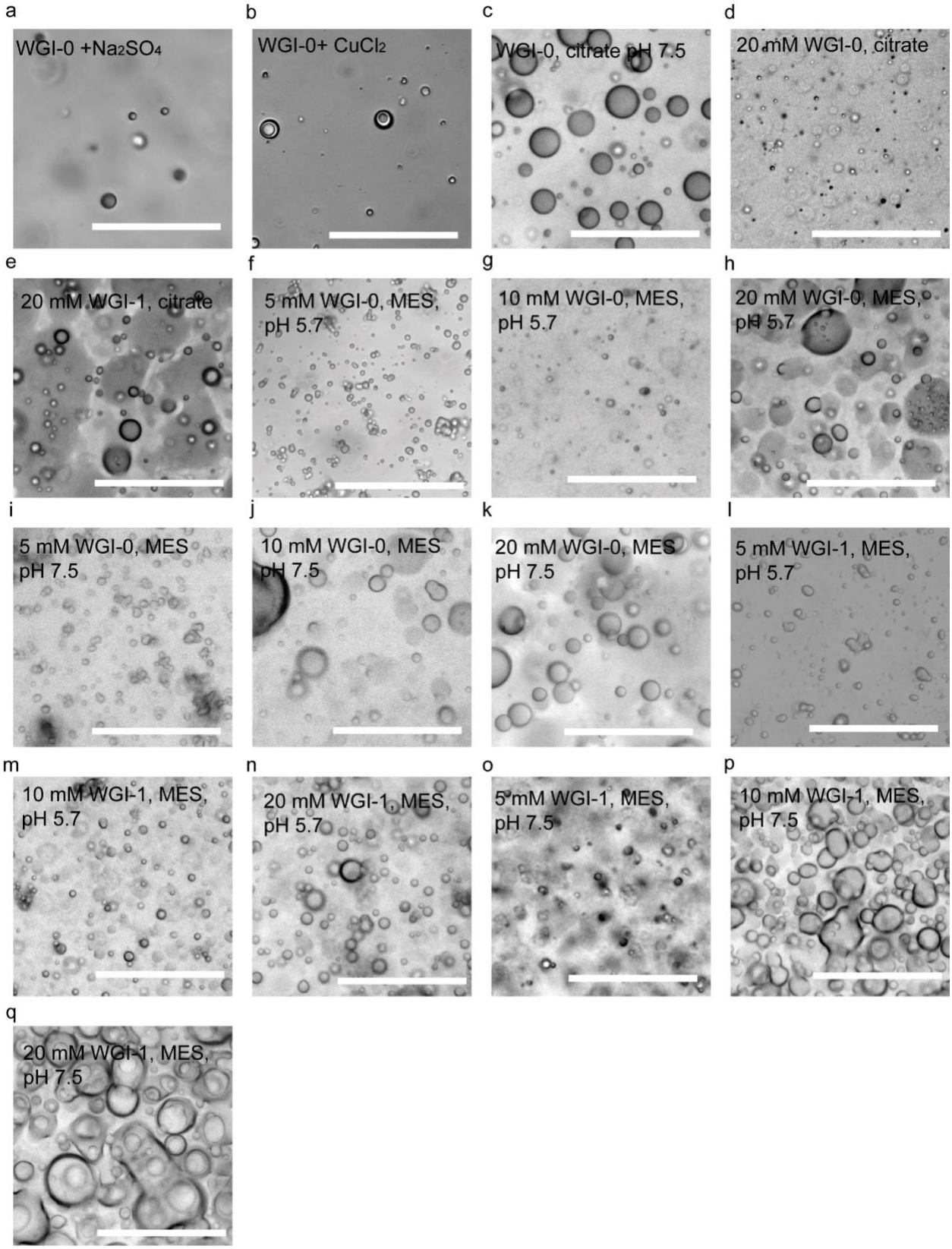
Supplementary Figure 7. Scheme of order of sample preparation of reactions in peptide condensates. i. Peptides were dissolved in citrate buffer at pH 5.5, or in MES buffer and the pH was adjusted to 7.5. ii. Reactants and NaAsc were added to the peptide solutions iii. After 30 minutes of Ar incubation. iv. $CuSO_4$ was added, resulting in both condensate formation and reaction initiation. v. The reaction was quenched following ACN addition.



Supplementary Figure 8. Calibration curve of the **a.** triazole ring product, **b.** AHC **c**. coumarin analyzed by HPLC.



Supplementary Figure 9. Microscopy analysis of 5mM WGI-2 condensate morphology in the presence of (**a**) NaAsc without the reactants and (**b**) the reactants without NaAsc. Scale bar= 50 μm.


a
WGI-0 +Na2SO4
b
WGI-0+ CuCl2
c
WGI-0, citrate pH 7.5
d
20 mM WGI-0, citrate
e
20 mM WGI-1, citrate
f
5 mM WGI-0, MES, pH 5.7
g
10 mM WGI-0, MES, pH 5.7
h
20 mM WGI-0, MES, pH 5.7
i
5 mM WGI-0, MES pH 7.5
j
10 mM WGI-0, MES pH 7.5
k
20 mM WGI-0, MES pH 7.5
l
5 mM WGI-1, MES, pH 5.7
m
10 mM WGI-1, MES, pH 5.7
n
20 mM WGI-1, MES, pH 5.7
o
5 mM WGI-1, MES, pH 7.5
p
10 mM WGI-1, MES, pH 7.5
q
20 mM WGI-1, MES, pH 7.5

Supplementary Figure 10. Different condition of WGI-0 and WGI-1. **a-b.** 10 mM WGI-0, citrate buffer, pH 5.5 with 10 mM **a.** Na_2SO_4, **b**. $CuCl_2$. **c**. 10 mM WGI-0, citrate buffer, pH 7.5, 10 mM $CuSO_4$. **d-e.** 20 mM, in citrate buffer pH 5.5, 10 mM $CuSO_4$ **d.** WGI-0, **e.** WGI-1. **f-h.** WGI-0 in MES buffer, pH 5.7, 10 mM $CuSO_4$ **f.** 5mM **g.** 10 mM **h.** 20 mM. **i-k.** WGI-0 in MES buffer, pH 7.5, 10 mM $CuSO_4$. **i.** 5mM **j.** 10 mM **k.** 20 mM **l-n.** WGI-1 in MES buffer, pH 5.7, 10 mM $CuSO_4$. **l.** 5mM **m.** 10 mM **n.** 20 mM **o-q.** WGI-1 in MES buffer, pH 7.5, 10 mM $CuSO_4$ **o.** 5mM **p.** 10 mM **q.** 20 mM. Scale bar= 50 μm.

Supplementary Table 1: *p*-values of figure 3b.

	p-value	<0.05 (*)	<0.01 (**)
buffer VS DDW	0.025900848	TRUE	FALSE
buffer VS WGI-0	0.002280834	TRUE	TRUE
buffer VS WGI-1	0.009654588	TRUE	TRUE
Buffer VS WGI-2	0.140036889	FALSE	FALSE
DDW VS WGI-0	0.511024099	FALSE	FALSE
DDW VS WGI-1	0.653099168	FALSE	FALSE
DDW vs WGI-2	0.035826605	TRUE	FALSE
WGI-0 VS WGI-1	0.779004664	FALSE	FALSE
WGI-0 VS WGI-2	0.002094457	TRUE	TRUE
WGI-1 VS WGI-2	0.011691681	TRUE	FALSE

Supplementary Table 2: *p*-values of figure 3e.

	p-value	<0.05 (*)
WGI-0 VS WGI-1	0.424721168	FALSE
WGI-0 vS buffer	0.083942996	FALSE
WGI-1 VS buffer	0.109652942	FALSE

Supplementary Table 3: *p*-values of figure 5b.

		<0.05 (*)	<0.01(**)
WGI-0 VS WGI-1	0.6379795	FALSE	FALSE
WGI-0 VS buffer	0.0315677	TRUE	FALSE
WGI-1 VS buffer	0.034747	TRUE	FALSE